# CSC 2114 Assignment 3: Weighted Campus Route Finder (UCS and IDS)

Report | Group <number>-<Day A/B/Eve> | Members: _____ | Program: `route_finder.py` (Python 3, standard library only)

## 1. Problem formulation (Part A)

| Component | Weighted formulation (type) | Changed since A1? |
|---|---|---|
| **1. Initial state** | $s_0 \in S$, the source school typed by the user. S is the set of 18 places (14 schools + 4 landmarks). An element of a set. | No |
| **2. Actions(s)** | Function S → finite set of actions. Actions(s) = {"walk to n" : n ∈ graph[s]}. Read off the adjacency dictionary `graph[s] = {neighbour: metres}` built from WEIGHTED_EDGES (both directions); listed in alphabetical order of n. The weight sits beside each action but does not decide which actions are legal. | Same form; the structure now carries weights |
| **3. Result(s,a)** | Function S × A → S. Result(s, "walk to n") = n. | No |
| **4. Goal test** | Boolean test on a state: s = destination (a school, never a landmark). Applied when a node is removed from the frontier. | No |
| **5. Path cost, step cost** | Path cost of a route = $\Sigma\, c(s_i, a_i, s_{i+1})$. Step cost is a function of the graph: c(s, a, s') = graph[s][s'] metres (S × A × S → positive integers, symmetric). It is positive because every weight is a positive integer; the smallest is 70 m, so ε = 70. | **Yes**: was the constant 1 |

**What the weights keep ignoring.** The model still ignores (i) slope (Makerere is hilly, so uphill and downhill legs of equal length are not equal effort), (ii) crowds and time of day (lecture changeovers, lunch queues), and (iii) rain, mud, flooding and road crossings. Slope can be folded in without changing the formulation: define c(s,a,s') = d(s,s')·(1 + k·uphill_gradient(s,s')), still a positive function of the graph, so UCS and the generic loop are untouched.

**Why positive step costs.** UCS needs every c ≥ ε > 0 so that path cost strictly grows along any path. With a weight of 0, a zero-cost cycle (A→B→A) lets a tree-like UCS regenerate nodes at the same cost forever, and ties make "cheapest first" meaningless, so it may never reach costlier nodes (non-termination). With a negative weight, a state already removed and expanded at cost g could later be reached more cheaply, so "first removal = cheapest" is false and UCS can return a non-optimal route; a negative cycle makes cost unbounded below and the search never settles.

**State versus node.** In my UCS trace from Law to the Built Environment, the state *Main Library* appears in two nodes: one reached via Economics with path cost 520 m (400 m edge) and one via Freedom Square with path cost 270 m. UCS should keep the 270 m node; the 520 m node is stale.

**What IDS needs from a node.** A depth-limited search needs the node's DEPTH (hops from the root). A node gets it from its parent when created: depth = parent.depth + 1, and 0 for the root.

**Which quantity is minimised.** BFS: hops, not metres. UCS: metres (total path cost). IDS: hops (it returns the first goal at the smallest depth limit, so like BFS it minimises hops and its cost in metres is incidental).

## 2. Design: what changed since Assignment 1

- **Graph and checker.** `build_graph` returns `dict[str, dict[str,int]]`; `neighbours(graph,s)` returns (neighbour, weight) pairs sorted by neighbour; `step_cost` reads one weight; `route_cost(graph, route)` re-adds the weights from the graph and an `assert` in `run_algorithm` checks it equals the printed cost for every route. All search code receives `graph` as an argument.
- **Node.** STATE, PARENT, ACTION, PATH-COST (metres from the root) and the new DEPTH = parent.depth + 1. Hops printed = DEPTH of the goal node. `solution()` is unchanged.
- **reached** now maps state → the cheapest node found so far. The update rule is a plug-in function: `never_replace` (BFS, DFS, IDS: first node kept) or `replace_if_cheaper` (UCS: replace when new.path_cost < old.path_cost, strictly).
- **Three frontier classes** with the same interface (push, pop, is_empty, __len__, plus push_order): `QueueFrontier` (deque, FIFO), `StackFrontier` (list, LIFO; push_order reverses the children so the alphabetically first pops first), `PriorityFrontier` (heapq of tuples (path_cost, state_name, counter, node)).
- **UCS tie-break and stale rule.** The heap tuple orders by lowest cost, then alphabetical state, then earliest insertion (the counter also stops Python comparing Nodes). A heap cannot delete from the middle, so a replaced node stays in the heap; when it is removed the loop sees `reached[node.state] is not node`, discards it silently and does not count it as expanded. With reached off nothing is ever discarded.
- **IDS driver.** `ids()` calls the same `search()` for limit = 0, 1, …, places−1 with a StackFrontier, `never_replace`, reached off, cycle check on. A node at depth = limit is still goal-tested but not expanded. `search` returns CUTOFF if a limit stopped some node and FAILURE if nothing was cut off (the whole space is explored, print "No route found"). Expansions are summed over iterations.
- **Weights never steer BFS, DFS or IDS.** They are read only in step_cost (to fill PATH-COST), in the heap ordering of UCS and in route_cost. Test: rebuilding the graph with every weight set to 1 gave identical routes *and* expansion counts for BFS, DFS and IDS on all 182 ordered school pairs.

Generic search loop (the frontier, the reached rule, the cycle check and the depth limit are the only plug-in points; there is no "if algorithm ==" branch):

```
search(graph, source, goal, frontier, replace_rule, use_reached, cycle_check, depth_limit, cap):
  root = Node(source);  frontier.push(root);  reached = {source: root}      # reached only if use_reached
  while frontier not empty:
      node = frontier.pop()                    # <- FRONTIER: queue / stack / priority queue (BFS / DFS+IDS / UCS)
      if use_reached and reached[node.state] is not node:  continue    # stale node: discard, not counted
      if node.state == goal:  return SOLUTION(node)                     # goal test on REMOVAL
      if depth_limit is not None and node.depth >= depth_limit:  cutoff = True;  continue   # <- DEPTH LIMIT (IDS)
      if cap reached:  return CAP
      expanded += 1                                                  # expand
      children = []
      for place in alphabetical neighbours of node.state:
          if cycle_check and place occurs on path root..node:  continue          # <- CYCLE CHECK (IDS)
          child = Node(place, node, place, node.path_cost + step_cost(graph, node.state, place))
          if use_reached and place in reached and not replace_rule(child, reached[place]):  continue  # <- REACHED RULE
          children.append(child)
      for child in frontier.push_order(children):                         # <- FRONTIER decides push order
          reached[child.state] = child;  frontier.push(child)
  return CUTOFF if cutoff else FAILURE
```

| Algorithm | Frontier | Reached rule | Cycle check / depth limit | reached |
|---|---|---|---|---|
| BFS | QueueFrontier | never_replace | none | on (off with --noreached) |
| DFS | StackFrontier | never_replace | none | on (off with --noreached) |
| UCS | PriorityFrontier | replace_if_cheaper | none | on (off with --noreached) |
| IDS (driver) | StackFrontier | never_replace | cycle check on, limit 0,1,2,… | off |

## 3. Verification (B4)

**3.1 Reference rows.** My program reproduces all three reference rows exactly (UCS cost / hops / expanded | IDS cost / hops / total expanded):

| Source → Destination | UCS | IDS | Matches? |
|---|---|---|---|
| School of Education → School of Economics | 550 / 4 / 8 | 560 / 2 / 4 | yes |
| CoCIS → School of Liberal and Performing Arts | 820 / 4 / 13 | 840 / 3 / 11 | yes |
| School of the Built Environment → School of Law | 870 / 5 / 11 | 1170 / 5 / 32 | yes |

**3.2 Watching reached change** (`--trace`, Law → Built Environment; full trace in Appendix B). Relevant lines:

```
[trace] remove School of Economics g=120
[trace] push Main Library g=520 (from School of Economics)
[trace] remove Freedom Square g=180
[trace] reached replace Main Library: g=520 -> g=270
[trace] push Main Library g=270 (from Freedom Square)
[trace] discard stale Main Library g=520 (reached holds g=270)
```

UCS removed Economics (g = 120) before Freedom Square (g = 180) because 120 < 180, and Economics reached Main Library over the long 400 m edge, so reached stored Main Library at 520. When Freedom Square was removed it reached Main Library over a 90 m edge at 270 < 520, so reached replaced the entry and a new node was pushed. The 520 node could not be pulled out of the heap; it stayed there and was discarded without expansion when it surfaced. The first node was not good enough because it was the *first* to reach the state, not the *cheapest*; the same thing happened to Margaret Trowell (1100 via Liberal → 920 via Engineering), and that replacement is what leads to the optimal 870 m route.

**3.3 Tree-like UCS.**

| Pair | UCS, reached on: expanded | UCS, --noreached: expanded | Max frontier (noreached) |
|---|---|---|---|
| School of Education → School of Economics | 8 (cost 550) | 35 (cost 550) | 87 |
| School of Statistics and Planning → School of Veterinary Medicine | 17 (cost 1840) | 10,000: **cap reached**, no route returned | 20,270 |

Without reached, every cycle in the graph (for example the triangle Main Building–Freedom Square–Main Library, or CoCIS–Central Teaching Facility–Statistics and Planning) is re-entered again and again, so the same state is regenerated through many routes and the frontier explodes (87 nodes for a 550 m trip, over 20,000 for the 1840 m trip). It would still stop if the cap were removed: every step costs at least ε = 70 m, so a node with path cost above $C^*$ needs more steps than $C^*/\varepsilon$ allows, only finitely many nodes have cost ≤ $C^*$, and once the cheapest goal node is at the front of the heap it is removed and returned. But for 1840 m that is up to 26 levels of branching, far beyond 10,000 expansions.

**3.4 IDS without reached.** It terminates because (a) the cycle check refuses any child whose state is already on the path from the root, so every path is simple and no longer than 17 hops, and (b) each iteration has a depth limit and the driver never goes above places − 1. What it gives up is memory of other branches: with no reached table it re-expands the same state every time a different route reaches it (and again in every iteration), so it does much more work (Section 5, Q4), and it cannot replace a node by a cheaper one.

## 4. Results (Part C)

The 40-row table (ten pairs × BFS, DFS, UCS, IDS; graph search, reached on, ALL mode) is in **Appendix A**. For pair *p* the rows are 4p−3 (BFS), 4p−2 (DFS), 4p−1 (UCS), 4p (IDS). Every cost was checked by the `route_cost` assertion, and BFS and DFS ignore the weights (Section 2), so their routes are the ones of Assignment 2. Conditions, with the rows that satisfy them (row of first algorithm vs row of second):

| | |
|---|---|
| UCS route cheaper than BFS (needs ≥ 3 pairs) | pairs 1 (3 vs 1), 2 (7 vs 5), 3 (11 vs 9), 4 (15 vs 13), 9 (35 vs 33), 10 (39 vs 37) |
| UCS route has more hops than BFS (≥ 1) | pairs 2 (7 vs 5), 3 (11 vs 9), 4 (15 vs 13), 9 (35 vs 33), 10 (39 vs 37) |
| IDS route costs more metres than UCS (≥ 2) | pairs 1 (4 vs 3), 2 (8 vs 7), 3 (12 vs 11), 4 (16 vs 15), 9 (36 vs 35), 10 (40 vs 39) |
| UCS expands more nodes than BFS (≥ 1) | pairs 3 (11 vs 9), 4 (15 vs 13), 7 (27 vs 25) |
| UCS expands fewer nodes than BFS (≥ 1) | pairs 1 (3 vs 1), 5 (19 vs 17), 10 (39 vs 37) |
| IDS expands ≥ 5× as many nodes as BFS (≥ 1) | pairs 2 (8 vs 5), 8 (32 vs 29), 9 (36 vs 33) |
| The two schools are directly connected (≥ 1) | pairs 5 (rows 17–20), 7 (rows 25–28) |
| Fewest-hops distance is 6 or more (≥ 1) | pairs 2 (6 hops, rows 5–8), 8 (9 hops, rows 29–32), 9 (8 hops, rows 33–36), 10 (9 hops, rows 37–40) |

**--verbose on the deepest pair** (pair 10, Veterinary Medicine → Built Environment: fewest-hops distance 9, UCS route has 10 hops, the most in the table; a second 9-hop pair, Engineering → Veterinary Medicine, is pair 8):

| Algorithm | Max frontier size | Cost (m) | Hops | Expanded |
|---|---|---|---|---|
| BFS | 4 | 2320 | 9 | 17 |
| DFS | 7 | 2080 | 10 | 12 |
| UCS | 5 | 2080 | 10 | 16 |
| IDS | 8 | 2320 | 9 | 76 |

IDS per iteration (limit: expanded): 0: 0, 1: 1, 2: 2, 3: 3, 4: 4, 5: 5, 6: 7, 7: 11, 8: 22, 9: 21; total 76, final depth limit 9.

**One terminal run of ALL** (Built Environment → Law; typed answers: *School of the Built Environment*, *School of Law*, *all*; the place menu is omitted):

```
Algorithm: BFS
Route: School of the Built Environment -> Margaret Trowell School of Industrial and Fine Art -> School of Liberal and
       Performing Arts -> Main Building -> Freedom Square -> School of Law
Cost (metres): 1170
Hops: 5
Nodes expanded: 12

Algorithm: DFS
Route: School of the Built Environment -> Margaret Trowell School of Industrial and Fine Art -> School of Liberal and
       Performing Arts -> Main Building -> Freedom Square -> School of Law
Cost (metres): 1170
Hops: 5
Nodes expanded: 9

Algorithm: UCS
Route: School of the Built Environment -> School of Engineering -> CoCIS -> Central Teaching Facility -> Freedom Square ->
       School of Law
Cost (metres): 870
Hops: 5
Nodes expanded: 11

Algorithm: IDS
Route: School of the Built Environment -> Margaret Trowell School of Industrial and Fine Art -> School of Liberal and
       Performing Arts -> Main Building -> Freedom Square -> School of Law
Cost (metres): 1170
Hops: 5
Nodes expanded: 32
Final depth limit: 5
```

## 5. Analysis

**Q1. UCS optimality.** The two conditions: (i) every step cost is at least some ε > 0 (here 70 m), so path cost strictly grows and nothing is stuck behind a zero or negative edge; (ii) the goal test is done when a node is *removed*, not when it is generated. When a node is removed it has the lowest path cost on the frontier, so no unexplored path can be cheaper; a node merely generated may have been reached by a poor path. Pair 3 (Education → Economics): UCS returns 550 m (Education → Main Library → Freedom Square → Law → Economics). A UCS that tests at generation expands Education, then Social Sciences (g = 140), then Main Library (g = 160), generates Economics there at 160 + 400 = 560 and returns at once: Education → Main Library → Economics at 560 m, 10 m dearer, because the 550 m route only reaches Economics later, after Law (g = 430) is expanded. I confirmed this with a generation-test variant of the loop.

**Q2. BFS stops being optimal.** Pair 1 (Built Environment → Law) in Figure 1. Both routes have 5 hops. BFS: Built → Margaret Trowell (70) → Liberal (450) → Main Building (350) → Freedom Square (120) → Law (180) = **1170 m**. UCS: Built → Engineering (90) → CoCIS (250) → Central Teaching Facility (130) → Freedom Square (220) → Law (180) = **870 m**. BFS only looks at depth: all 5-hop routes are equal to it, so it keeps the first one that reaches Law in FIFO order, and the Margaret Trowell branch is queued first (alphabetical order), even though it uses the 450 m and 350 m edges. Pair 2 shows the other case: UCS is cheaper (1580 vs 1790 m) with *more* hops (8 vs 6).

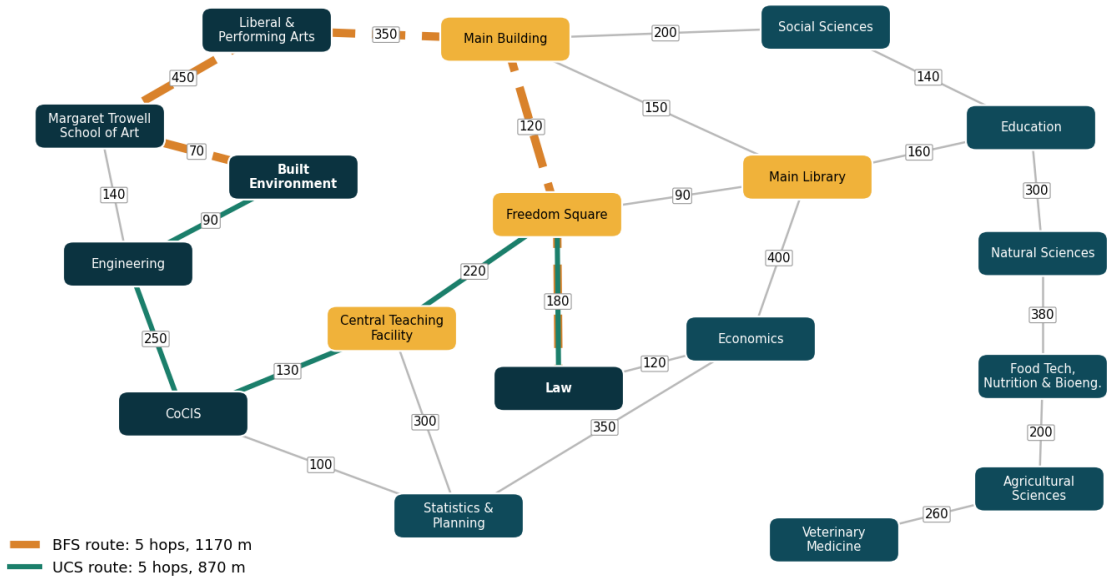



*Figure 1. Built Environment → Law on the weighted campus graph: BFS (dashed orange) versus UCS (solid green). Grey edges are not used.*

**Q3. IDS versus BFS and UCS.** IDS is never cheaper than UCS (UCS is optimal). IDS's cost equals UCS's when the fewest-hops route is also the cheapest: pairs 5, 6, 7, 8 (e.g. pair 8: 1990 m for both; pair 5: 100 m). It is dearer when a longer-in-hops detour is shorter in metres: pair 2 (+210 m: 1790 vs 1580), pair 10 (+240 m), pair 1 (+300 m), pair 9 (+120 m), pair 3 (+10 m). In all ten pairs IDS's cost equals BFS's, since both return the first fewest-hops route in alphabetical order.

**Q4. Where IDS's repeated work goes.** Pair 10: expansions at limits 0 to 9 are 0, 1, 2, 3, 4, 5, 7, 11, 22, 21; total **76**, against 17 for BFS (4.5×); pair 8 is worse (433 vs 17, with 183 at limit 8 alone). On a tree with a large branching factor b the argument holds because the last level has about $b^d$ nodes and all shallower levels together are only a fraction $1/(b-1)$ of it, so repeating them adds roughly a factor $(b/(b-1))^2$ (about 1.2 for b = 10). On this graph it fails: it has only 18 states and average degree 2.7, so the levels do not grow geometrically, and with no reached table IDS counts *simple paths*, not states. The number of simple paths of length ≤ L through the cyclic core (Main Building, Freedom Square, Main Library, Central Teaching Facility, CoCIS, Statistics and Planning, Economics, Law) grows much faster than the 17 states BFS expands once each, and the shallow iterations (55 of the 76 expansions) repeat all of it.

**Q5. Memory.** Max frontier size, pair 10: BFS 4 < UCS 5 < DFS 7 < IDS 8 (IDS: largest over its iterations). This is the opposite of the textbook ranking, and it comes from the graph: the route from Veterinary Medicine through Agricultural Sciences, Food Technology and Natural Sciences to Education is a chain of degree-2 places, so BFS's frontier stays tiny until the wave enters the dense core (Freedom Square has degree 4, Main Library 4), where each expansion adds up to 3 nodes. BFS and UCS use reached, so each state is queued about once and the frontier can never exceed the 18 states; UCS is one larger because its heap also holds stale entries (Main Building 1480, Economics 1700, Statistics and Planning 1910 were replaced and later discarded). DFS and IDS have no such bound: the stack keeps every unexplored sibling along the current path (up to 9 deep), and IDS has no reached table. The lecture's claim is that IDS needs memory linear in the depth (O(bd)) whereas BFS needs $O(b^d)$; with only 18 states, BFS's exponential bound is capped by the state count, so the asymptotic advantage of IDS does not show here.

**Q6. Two ways to avoid loops.** *Memory:* reached stores one node per state ever generated (at most 18 entries here, growing with the state space); the cycle check stores nothing extra, it walks the parent chain of the current path, so it needs O(depth) time per check and no table. *Reopening:* only reached with the replace-if-cheaper rule lets UCS reopen a state at a lower cost (the Main Library 520 → 270 replacement in Section 3.2); the cycle check forbids only a state on the node's own ancestry, so it never compares costs. *What each can wrongly discard:* reached with `never_replace` throws away every later path to a known state, even a cheaper one (harmless for BFS's hops, but it is why DFS can return a dear route); with a depth limit it could even discard a shallower path to a state first seen deeper, which is why IDS does not use it. The cycle check discards only routes that return to a state on their own path; with positive costs such a route can never be better than the path without the loop, so nothing useful is lost, but non-loop repeats (two different routes into the same state) are not caught, which costs time.

**Q7. Do the uninformed algorithms read the weights?** Prediction for Margaret Trowell–Liberal dropping from 450 m to 150 m: BFS, DFS and IDS never use weights to choose, so their *routes* and expansion counts cannot change; but a route that uses that edge is reported with a lower cost (cost is recomputed from the graph). UCS reads the weights, so it can change both route and cost. Only pairs whose routes avoid that edge are unaffected by all four. Check, over all 182 ordered school pairs: BFS 0 route changes (38 cost-only changes), DFS 0 (25), IDS 0 (38), expansion counts of BFS, DFS, IDS unchanged in all cases; UCS changed its route in 30 pairs and the cost of an unchanged route in 8. Example, CoCIS → Liberal: BFS and IDS keep CoCIS → Engineering → Margaret Trowell → Liberal and its cost falls 840 → 540; UCS switches from its 4-hop 820 m route via Freedom Square and Main Building to that same 3-hop route at 540 m (expansions 13 → 11). The prediction held.

**Q8. Limitation of UCS.** UCS expands every place cheaper than the goal, in every direction. Pair 6 (CoCIS → Social Sciences): UCS made 12 expansions, and 4 of them moved away from the destination, that is, the place was farther from Social Sciences (true remaining metres) than the start was (670 m): Statistics and Planning (770), Engineering (920), Margaret Trowell (1000) and Built Environment (1010). Over all 182 ordered pairs the worst is Education → Veterinary Medicine, with 13 of 17 expansions going the wrong way. **A*** (from the lecture's best-first table; greedy best-first also uses a heuristic but is not optimal) would avoid most of this. It needs a heuristic h(n), an estimate of the remaining distance to the destination that never overestimates (admissible,

ideally consistent), for example the straight-line distance in metres from map coordinates of every place, which this model does not hold yet; A* orders the frontier by g(n) + h(n).

## 6. Reflection

The change I would make is to let the walking cost depend on direction: walking uphill from a lower school to a higher one takes longer than walking down, so the symmetric weight c(A,B) = c(B,A) is unrealistic on Makerere's hills. This changes only component 5 (the step cost becomes asymmetric, c(s,a,s') ≠ c(s',a',s)), plus the way the graph is stored (a directed adjacency dictionary); the initial state, actions, result and goal test stay the same, and UCS remains correct as long as every cost stays ≥ ε. A time-of-day cost would be bigger: the state would have to include the clock time, which changes the state space, the initial state and Result as well.

## Appendix A: results table (40 rows, graph search, reached on)

Place abbreviations: MainBldg = Main Building; FreedomSq = Freedom Square; MainLib = Main Library; CTF = Central Teaching Facility; Liberal = School of Liberal and Performing Arts; SocSci = School of Social Sciences; MargTrowell = Margaret Trowell School of Industrial and Fine Art; BuiltEnv = School of the Built Environment; Eng = School of Engineering; Educ = School of Education; NatSci = College of Natural Sciences; FoodTech = School of Food Technology, Nutrition and Bioengineering; Agri = School of Agricultural Sciences; Vet = School of Veterinary Medicine; CoCIS = CoCIS; Law = School of Law; Econ = School of Economics; StatPlan = School of Statistics and Planning.

| # | Source | Destination | Algorithm | Route | Cost (m) | Hops | Expanded |
|---|---|---|---|---|---|---|---|
| 1 | BuiltEnv | Law | BFS | BuiltEnv → MargTrowell → Liberal → MainBldg → FreedomSq → Law | 1170 | 5 | 12 |
| 2 | BuiltEnv | Law | DFS | BuiltEnv → MargTrowell → Liberal → MainBldg → FreedomSq → Law | 1170 | 5 | 9 |
| 3 | BuiltEnv | Law | UCS | BuiltEnv → Eng → CoCIS → CTF → FreedomSq → Law | 870 | 5 | 11 |
| 4 | BuiltEnv | Law | IDS | BuiltEnv → MargTrowell → Liberal → MainBldg → FreedomSq → Law | 1170 | 5 | 32 |
| 5 | StatPlan | Agri | BFS | StatPlan → Econ → MainLib → Educ → NatSci → FoodTech → Agri | 1790 | 6 | 16 |
| 6 | StatPlan | Agri | DFS | StatPlan → CTF → FreedomSq → MainBldg → SocSci → Educ → NatSci → FoodTech → Agri | 1860 | 8 | 12 |
| 7 | StatPlan | Agri | UCS | StatPlan → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri | 1580 | 8 | 16 |
| 8 | StatPlan | Agri | IDS | StatPlan → Econ → MainLib → Educ → NatSci → FoodTech → Agri | 1790 | 6 | 154 |
| 9 | Educ | Econ | BFS | Educ → MainLib → Econ | 560 | 2 | 7 |
| 10 | Educ | Econ | DFS | Educ → MainLib → Econ | 560 | 2 | 16 |
| 11 | Educ | Econ | UCS | Educ → MainLib → FreedomSq → Law → Econ | 550 | 4 | 8 |
| 12 | Educ | Econ | IDS | Educ → MainLib → Econ | 560 | 2 | 4 |
| 13 | CoCIS | Liberal | BFS | CoCIS → Eng → MargTrowell → Liberal | 840 | 3 | 11 |
| 14 | CoCIS | Liberal | DFS | CoCIS → CTF → FreedomSq → MainBldg → Liberal | 820 | 4 | 4 |
| 15 | CoCIS | Liberal | UCS | CoCIS → CTF → FreedomSq → MainBldg → Liberal | 820 | 4 | 13 |
| 16 | CoCIS | Liberal | IDS | CoCIS → Eng → MargTrowell → Liberal | 840 | 3 | 11 |
| 17 | CoCIS | StatPlan | BFS | CoCIS → StatPlan | 100 | 1 | 3 |
| 18 | CoCIS | StatPlan | DFS | CoCIS → StatPlan | 100 | 1 | 17 |
| 19 | CoCIS | StatPlan | UCS | CoCIS → StatPlan | 100 | 1 | 1 |
| 20 | CoCIS | StatPlan | IDS | CoCIS → StatPlan | 100 | 1 | 1 |
| 21 | CoCIS | SocSci | BFS | CoCIS → CTF → FreedomSq → MainBldg → SocSci | 670 | 4 | 12 |
| 22 | CoCIS | SocSci | DFS | CoCIS → CTF → FreedomSq → MainBldg → SocSci | 670 | 4 | 7 |
| 23 | CoCIS | SocSci | UCS | CoCIS → CTF → FreedomSq → MainBldg → SocSci | 670 | 4 | 12 |
| 24 | CoCIS | SocSci | IDS | CoCIS → CTF → FreedomSq → MainBldg → SocSci | 670 | 4 | 19 |
| 25 | MargTrowell | Liberal | BFS | MargTrowell → Liberal | 450 | 1 | 2 |
| 26 | MargTrowell | Liberal | DFS | MargTrowell → Liberal | 450 | 1 | 16 |
| 27 | MargTrowell | Liberal | UCS | MargTrowell → Liberal | 450 | 1 | 4 |
| 28 | MargTrowell | Liberal | IDS | MargTrowell → Liberal | 450 | 1 | 1 |
| 29 | Eng | Vet | BFS | Eng → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri → Vet | 1990 | 9 | 17 |
| 30 | Eng | Vet | DFS | Eng → CoCIS → CTF → FreedomSq → MainBldg → SocSci → Educ → NatSci → FoodTech → Agri → Vet | 2200 | 10 | 11 |
| 31 | Eng | Vet | UCS | Eng → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri → Vet | 1990 | 9 | 17 |
| 32 | Eng | Vet | IDS | Eng → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri → Vet | 1990 | 9 | 433 |
| 33 | MargTrowell | Vet | BFS | MargTrowell → Liberal → MainBldg → MainLib → Educ → NatSci → FoodTech → Agri → Vet | 2250 | 8 | 17 |
| 34 | MargTrowell | Vet | DFS | MargTrowell → Eng → CoCIS → CTF → FreedomSq → MainBldg → SocSci → Educ → NatSci → FoodTech → Agri → Vet | 2340 | 11 | 11 |
| 35 | MargTrowell | Vet | UCS | MargTrowell → Eng → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri → Vet | 2130 | 10 | 17 |
| 36 | MargTrowell | Vet | IDS | MargTrowell → Liberal → MainBldg → MainLib → Educ → NatSci → FoodTech → Agri → Vet | 2250 | 8 | 324 |
| 37 | Vet | BuiltEnv | BFS | Vet → Agri → FoodTech → NatSci → Educ → MainLib → MainBldg → Liberal → MargTrowell → BuiltEnv | 2320 | 9 | 17 |
| 38 | Vet | BuiltEnv | DFS | Vet → Agri → FoodTech → NatSci → Educ → MainLib → FreedomSq → CTF → CoCIS → Eng → BuiltEnv | 2080 | 10 | 12 |
| 39 | Vet | BuiltEnv | UCS | Vet → Agri → FoodTech → NatSci → Educ → MainLib → FreedomSq → CTF → CoCIS → Eng → BuiltEnv | 2080 | 10 | 16 |
| 40 | Vet | BuiltEnv | IDS | Vet → Agri → FoodTech → NatSci → Educ → MainLib → MainBldg → Liberal → MargTrowell → BuiltEnv | 2320 | 9 | 76 |

## Appendix B: full `--trace` for UCS, School of Law → School of the Built Environment

```
[trace] remove School of Law g=0
[trace] push Freedom Square g=180 (from School of Law)
[trace] push School of Economics g=120 (from School of Law)
[trace] remove School of Economics g=120
[trace] push Main Library g=520 (from School of Economics)
[trace] push School of Statistics and Planning g=470 (from School of Economics)
[trace] remove Freedom Square g=180
[trace] reached replace Main Library: g=520 -> g=270
[trace] push Central Teaching Facility g=400 (from Freedom Square)
[trace] push Main Building g=300 (from Freedom Square)
[trace] push Main Library g=270 (from Freedom Square)
[trace] remove Main Library g=270
[trace] push School of Education g=430 (from Main Library)
[trace] remove Main Building g=300
[trace] push School of Liberal and Performing Arts g=650 (from Main Building)
[trace] push School of Social Sciences g=500 (from Main Building)
[trace] remove Central Teaching Facility g=400
[trace] push CoCIS g=530 (from Central Teaching Facility)
[trace] remove School of Education g=430
[trace] push College of Natural Sciences g=730 (from School of Education)
[trace] remove School of Statistics and Planning g=470
[trace] remove School of Social Sciences g=500
[trace] discard stale Main Library g=520 (reached holds g=270)
[trace] remove CoCIS g=530
[trace] push School of Engineering g=780 (from CoCIS)
[trace] remove School of Liberal and Performing Arts g=650
[trace] push Margaret Trowell School of Industrial and Fine Art g=1100 (from School of Liberal and Performing Arts)
[trace] remove College of Natural Sciences g=730
[trace] push School of Food Technology, Nutrition and Bioengineering g=1110 (from College of Natural Sciences)
[trace] remove School of Engineering g=780
[trace] reached replace Margaret Trowell School of Industrial and Fine Art: g=1100 -> g=920
[trace] push Margaret Trowell School of Industrial and Fine Art g=920 (from School of Engineering)
[trace] push School of the Built Environment g=870 (from School of Engineering)
[trace] remove School of the Built Environment g=870
```